

Sample estimate,

Sample mean of population Parameter Mean (μ):

The sample mean:

$$\bar{X} = \frac{\sum X}{n}$$

is a sample estimator (unbiased)
of the population parameter
mean (μ)

Example:

2, 5, 7

$$\bar{X} = \frac{2+5+7}{3} = \frac{14}{3} = 4.67.$$

Sample estimation of population Parameter Variance (σ^2) and

S.D σ :

$$\text{Diff of squares} = \frac{\sum X^2 - \frac{(\sum X)^2}{N}}{N}$$

mean of squares / Square of means.

DAY: _____

DATE: _____

Sample Variance (S^2)

$$S^2 = \frac{\sum (X - \bar{X})^2}{n}$$

is the biased (or standard)
estimator of population variance.
(σ^2)

Population

$$\text{Variance} = \sigma^2 = \frac{\sum (X - \bar{X})^2}{N-1}$$

Sample Standard Deviation (S).

$$S = \sqrt{\frac{\sum (X - \bar{X})^2}{n-1}} = \sqrt{\frac{\sum X^2 - \frac{(\sum X)^2}{n}}{n-1}}$$

Unbiased estimator of S.D (σ).
Represents the square root of
variance.

Measure dispersion of data ($X - \bar{X}$).
Based on mean of squared
deviations.

$$s^2 = \frac{1}{n-1} \left[\sum x^2 - \frac{(\sum x)^2}{n} \right]$$

s^2 is the estimated unbiased estimator of population variance (σ^2).